

Math 1232 Summer 2026

Midterm Exam

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, one-sided cheat sheet you have made for yourself ahead of time.
- You may use a four-function calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- The exam has 10 problems, and is 6 pages total.
- Each problem is worth ten points. The whole test is scored out of 100 points.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

1	2	3	4	5
6	7	8	9	10

Total /100

Name:

Name: _____

Problem 1. Compute $\int \frac{1}{x(2 + \ln(x))} dx$.

Problem 2. Compute $\frac{d}{dx} e^{\arccos(x^2)}$.

Problem 3. Let $h(x) = x^5 + x$. Compute $(h^{-1})'(2)$.

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Problem 4. Compute $\lim_{x \rightarrow \infty} x^{\frac{3}{x+2}}$.

Problem 5. Compute $\int x e^{3x} dx$.

Name: _____

Problem 6. Compute $\int \frac{1}{x^2 \sqrt{4+x^2}} dx$.

Name: _____

Problem 7. Compute $\int \frac{10}{(x+2)(x-3)} dx$.

Problem 8. Compute $\int \sin^3(x) \cos^6(x) dx$.

Name: _____

Problem 9. Compute $\int_1^\infty \frac{\ln(x)}{x} dx$. Does the integral converge or diverge?

Problem 10. Set up (but don't compute!) an integral for the area of a surface obtained by taking the curve $y = \ln(x^3 + 1)$ from $x = 0$ to $x = 10$ and rotating around the x -axis.